ECEN 603 Project:

Design of a Traffic Light Controller Using Behavioral Verilog HDL

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Introduction

- Traffic congestion is a growing problem in urban areas which leads to delays, emissions, accidents.
- Traditional traffic lights use fixed timing with no adaptability.
- Smart traffic systems can respond to real-time conditions, making them better at managing traffic.
- Hardware-based controllers (FPGAs, ASICs) offer deterministic, low-latency signal control.
- This project implements a multi-mode traffic light controller in synthesizable Verilog HDL

Background

- Zhilevski et al. [1] implemented a traffic light system in Verilog for a two-road intersection.
- Their approach: gate-level primitives (buf, not, and, or), no explicit state encoding.
- Separate source files for each mode, no top-level wrapper, no runtime switching.
- Our approach: behavioral RTL state machines, explicit state encoding, parameterized timing, unified top-level mux with mode_sel input, guaranteed latch-free.

Intersection Overview

- Two perpendicular roads, two lanes each, 8 traffic lights total.
- T11/T12 (vehicle, road 1), T21/T22 (vehicle, road 2), P11/P12/P21/P22 (pedestrian).
- Paired lights on same road are identical and thus the controller drives one set per road.
- Three modes: Normal (50/50, 80s cycle), Increased Traffic (asymmetric split – can be 75/25 or 62.5/37.5), Emergency (flashing yellow).

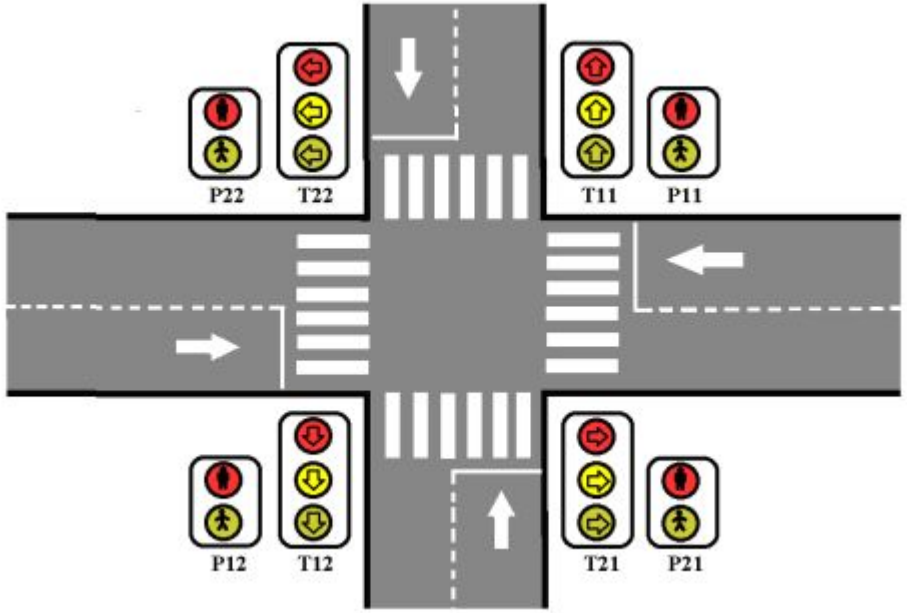


Figure 1: The intersection from Zhilevski et al. [1]

Normal Mode FSM & ASM Chart

- 5 vehicle phases: Red -> Red+Yellow -> Green -> Flashing Green -> Yellow.
  - 5-state Moore machine
- 80-second cycle, equal split between roads.
- Pedestrians walk when their road's cars are stopped.
- Flashing green warns drivers before yellow.

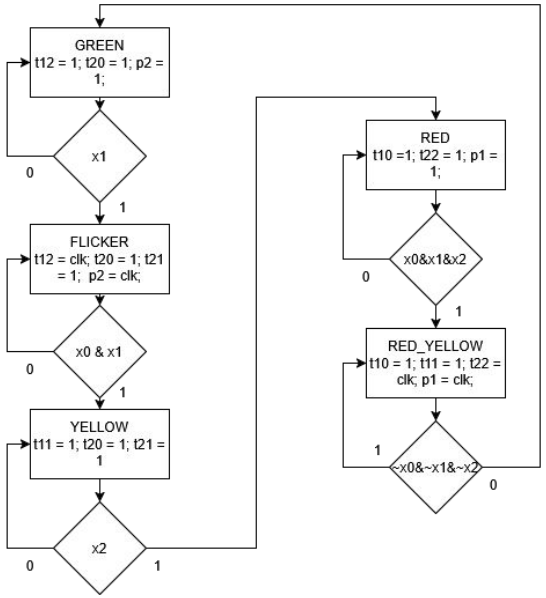


Figure 2: ASM Chart of Normal Operation

Increased Traffic Mode FSM & ASM Chart

- 6-state Moore machine: T1 Red+Yellow -> T1 Green -> T1 Yellow -> T2 Red+Yellow -> T2 Green -> T2 Yellow.
- RY and Yellow fixed at 2s, green durations are variable.
- 75/25 split: 56s/16s green, 62.5/37.5 split: 46s/26s green.
- Both pedestrian signals red during R+Y transitions for safety

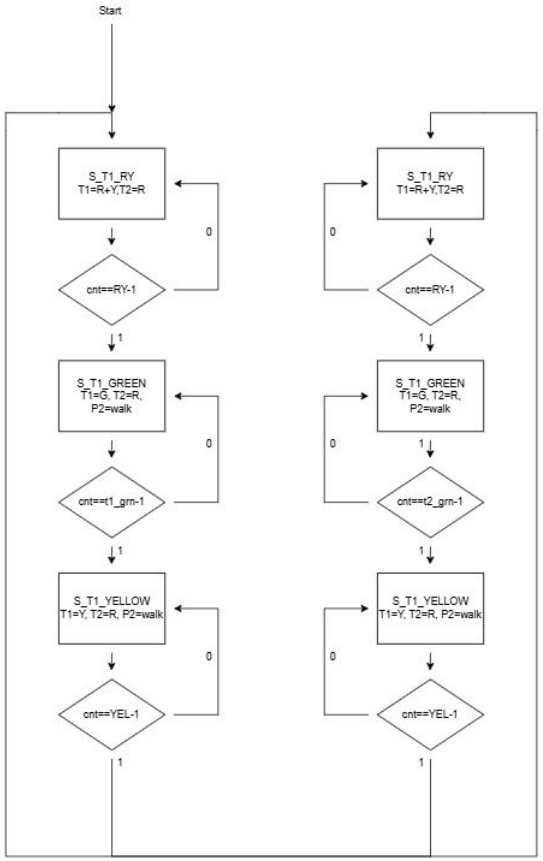


Figure 3: ASM Chart of Increased Traffic Operation

Increased Traffic Duration Table

Mode	xl (priority)	T1 Green	T2 Green	T1 Red %	T2 Red %
0 (75/25)	0 (T1 Priority)	56s	16s	25%	75%
0 (75/25)	1 (T2 Priority)	16s	56s	75%	25%
1 (62.5/37.5)	0 (T1 Priority)	46s	26s	37.5%	62.5%
1 (62.5/37.5)	1 (T2 Priority)	26s	46s	62.5%	37.5%

Emergency Mode

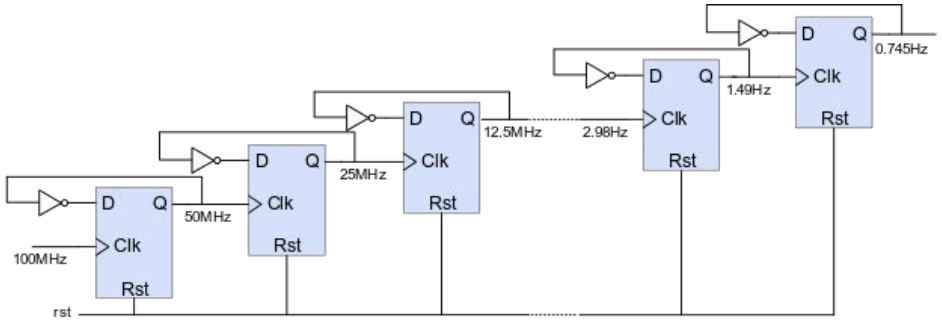
- No FSM needed.
- Counter toggles T1 and T2 yellow every 2 clock cycles.
- All other outputs (red, green, pedestrian) held low.
- When reset, counter = 0, yellow starts off.
- Used when system is disabled, malfunctioning, or during low-traffic nighttime hours.

Top-Level Controller

Houses Cascading Clock Delays (x0, x1, x2)

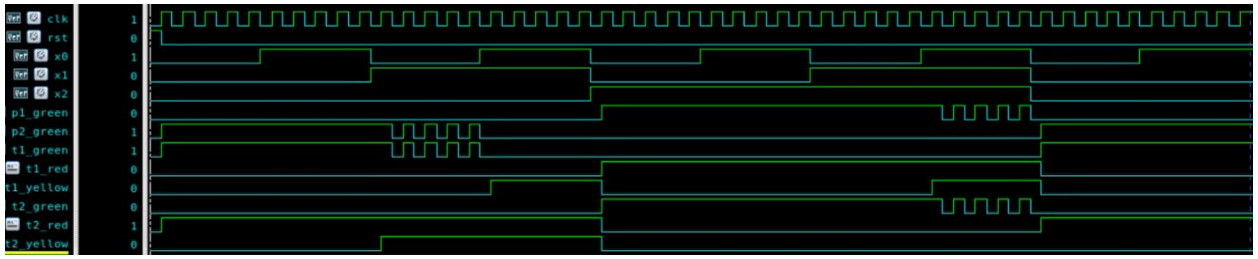
Mux to select between different output modes or modules

An Autonomous mode which selects between Normal, Low Priority Traffic, and High Priority Traffic based on duration of road priority

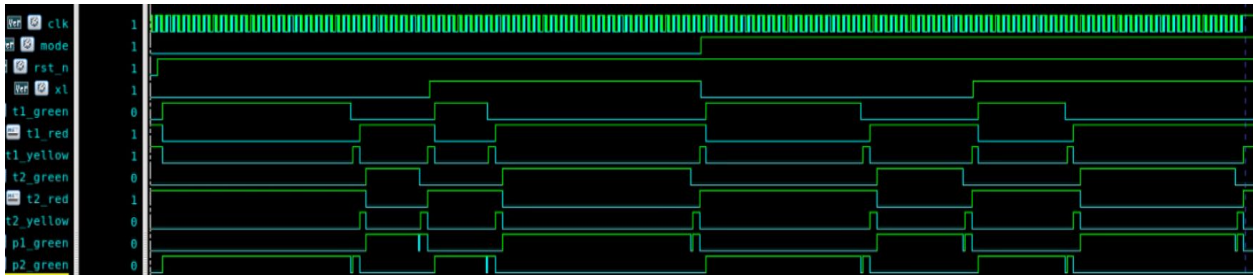



```
always @ (negedge clk40 or posedge rst) begin
    if (rst) count <= 3'b000;
    else begin
        if(x1) begin
            count <= count == 3'b011 || (hold_a & ~mode) ? count : count +1;
        end
        else begin
            count <= count == 3'b101 || (hold_a & ~mode) ? count : count-1;
        end
    end
end
```

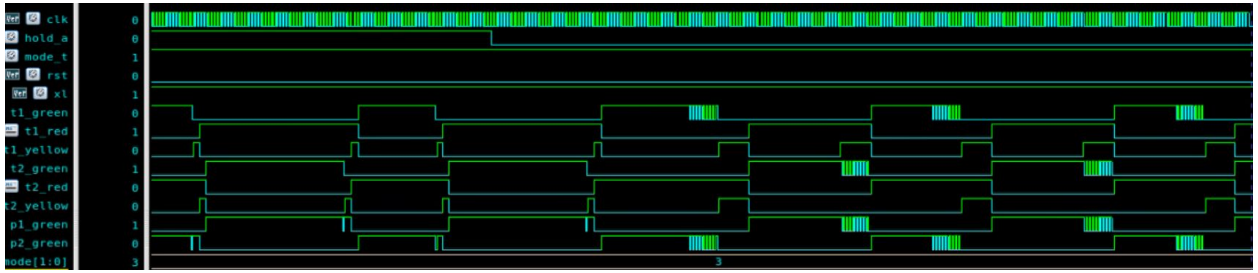
Simulation Results



Normal - Equal timing



Traffic - Priority swaps



Auto - Mode switching

Synthesis Results

Main constraint was Traffic Mode

- Critical path for Timing
- 40% of Power consumption
- 55% of Area

Another big cost was Clock Network (Normal control inputs)

Overall, still synthesizable with Top level module adding minimal overhead

- Timing isn't restricted
- 25% of Area
- Significant Power draw due to clock network

```
Number of ports:                          50
Number of nets:                          339
Number of cells:                         273
Number of combinational cells:           229
Number of sequential cells:               34
Number of macros/black boxes:              0
Number of buf/inv:                        50
Number of references:                     27

Combinational area:               413.000000
Buf/Inv area:                      54.000000
Noncombinational area:            284.000000
Macro/Black Box area:               0.000000
Net Interconnect area:      undefined  (No wire load specified)

Total cell area:                  697.000000
Total area:                 undefined
```

```
Point                                       Incr       Path
-----------------------------------------------------------
clock clk' (rise edge)                      5.00       5.00
clock network delay (ideal)                 0.00       5.00
clk10_reg/CP (FDS2)                         0.00       5.00 r
clk10_reg/Q (FDS2)                          1.40       6.40 f
dut_n/x1 (normal)                           0.00       6.40 f
dut_n/U9/Z (IVA)                            0.31       6.71 r
dut_n/U12/Z (AN4P)                          1.23       7.94 r
dut_n/current_state_reg[0]/D (FD1)          0.00       7.94 r
data arrival time                                      7.94

clock clk (rise edge)                      10.00      10.00
clock network delay (ideal)                 0.00      10.00
dut_n/current_state_reg[0]/CP (FD1)         0.00      10.00 r
library setup time                         -0.80       9.20
data required time                                     9.20
-----------------------------------------------------------
data required time                                     9.20
data arrival time                                     -7.94
-----------------------------------------------------------
slack (MET)                                            1.26
```

```
                 Internal         Switching          Leakage            Total
Power Group      Power            Power              Power              Power
 (   %    )  Attrs
--------------------------------------------------------------------------------
-----------------
io_pad             0.0000           0.0000            0.0000               NA
(     N/A)
memory             0.0000           0.0000            0.0000               NA
(     N/A)
black_box          0.0000           0.0000            0.0000               NA
(     N/A)
clock_network      0.0000        2.4016e+03           0.0000               NA
(     N/A)  i
register           0.0000        2.4252e+03           0.0000               NA
(     N/A)
sequential         0.0000          855.1407           0.0000               NA
(     N/A)
combinational      0.0000        6.1493e+03           0.0000               NA
(     N/A)
--------------------------------------------------------------------------------
-----------------
Total              0.0000 nW     1.1831e+04 nW        0.0000               NA
```

Controller:
Area
Timing
Power

Real-World Applicability & Improvements

- Real intersections must transition between modes without resetting hardware, and Zhilevski's design cannot do this.
- Parameterized timing lets engineers adjust ratios by changing constants, not redesigning logic.
- Emergency fallback to flashing yellow is a real regulatory requirement.

Limitations

- This traffic light system is not very practical in the real world, as it only handles straight-through traffic and does not account for turning movements.
- A truly adaptive system would need real-time sensor feedback and dynamic timing, not fixed ratios, but the modular top-level structure does make this possible.

Conclusion

- Explicit behavioral FSMs replace implicit gate-level states.
- Parameterized timing; change localparams, not logic.
- Guaranteed latch-free via default assignments.
- Unified top-level controller with runtime mode switching.
- Single increased traffic module replaces two separate files from [1].
- Inherently serial; FSM cannot be pipelined, and parallelism is irrelevant when only one mode active at a time.

References

[1] M. Zhilevski, M. Mikhov, and M. Zhilevska, "Design of a Traffic Light System Using Verilog HDL," in 2023 5th International Congress on Human-Computer Interaction, Optimization and Robotic Applications (HORA), Istanbul, Turkiye, 2023, pp. 1-6, doi: 10.1109/HORA58378.2023.10156804.